md2pdf Comprehensive Demo

Rust Rewrite Feature Showcase

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Open Source Project

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Introduction

This document demonstrates the features implemented in the **md2pdf** Rust rewrite. The tool converts Markdown to PDF using Typst as the rendering engine, eliminating the need for Chromium/Puppeteer.

Key Benefits

- **Fast conversion**: Sub-second PDF generation
- **Small binary**: No Chromium dependency (20MB vs 500MB)
- **Native rendering**: Uses Typst for high-quality output

Text Formatting

Basic Formatting

Regular text with *italic*, **bold**, and ***bold italic*** styles.

You can also use `inline code` for technical terms like `fn main()` or `cargo build`.

Here's some ~~strikethrough~~ text for corrections.

Links

Visit the Typst documentation for more information.

Lists

Unordered Lists

- First item
- Second item - Nested item A
  - Nested item B - Deeply nested
- Third item

Ordered Lists

1. First step
2. Second step
3. Third step + Sub-step A
   1. Sub-step B

Task Lists

- [x] Project setup with Cargo
- [x] CLI argument parsing with Clap
- [x] Markdown parsing with pulldown-cmark
- [x] Typst conversion
- [x] PDF generation
- [ ] Web API (optional feature)
- [ ] MCP Server (optional feature)

Code Blocks

Rust Example

```rust
use anyhow::Result;
use clap::Parser;

#[derive(Parser)]
struct Args {
    /// Input markdown file
    input: PathBuf,

    /// Output PDF file
    #[arg(short, long)]
    output: Option<PathBuf>,
}

fn main() -> Result<()> {
    let args = Args::parse();
    println!("Converting {:?} to PDF...", args.input);
    Ok(())
}
```

TypeScript Example

```typescript
interface DocumentMetadata {
  title?: string;
  subtitle?: string;
  author?: string;
  date?: string;
  toc?: boolean;
  cover?: boolean;
}

async function convertMarkdown(
  content: string,
  metadata: DocumentMetadata
): Promise<Uint8Array> {
  // Convert markdown to PDF
  return new Uint8Array();
}
```

Python Example

```python
from dataclasses import dataclass
from typing import Optional

@dataclass
class Config:
    input_file: str
    output_file: Optional[str] = None
    font: str = "Inter"
    font_size: int = 11
```

```python
def convert_to_pdf(config: Config) -> bytes:
    """Convert markdown file to PDF."""
    with open(config.input_file) as f:
        content = f.read()
    # ... conversion logic
    return b""
```

Tables

Feature Comparison

Binary Size	500MB	20MB
Startup Time	2-5 seconds	<100ms
Memory Usage	200MB	20MB
Dependencies	Chromium, Node.js	None
Math Support	KaTeX	Native

Implementation Status

CLI	Complete	All options implemented
Frontmatter	Complete	YAML parsing
Markdown	Complete	Full CommonMark
Code Blocks	Complete	With titles
Tables	Complete	With alignment
Math	Complete	LaTeX to Typst
Callouts	Complete	GitHub-style
Directives	Complete	Page/column layouts
Mermaid	Complete	Via kroki.io
Web API	Not Started	Future work

Math Equations

Inline Math

The famous equation $E = mc^2$ shows mass-energy equivalence.

Greek letters: α, β, γ, δ, π, ω.

Display Math

The Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

Euler's identity:

$$e^{i\pi} + 1 = 0$$

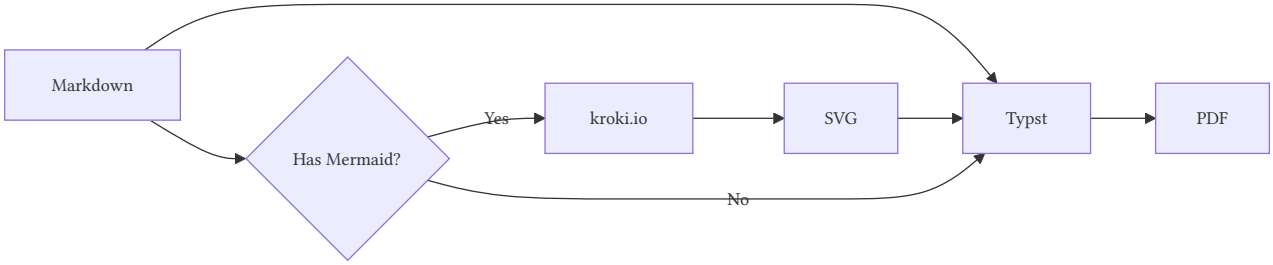
A series:

$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

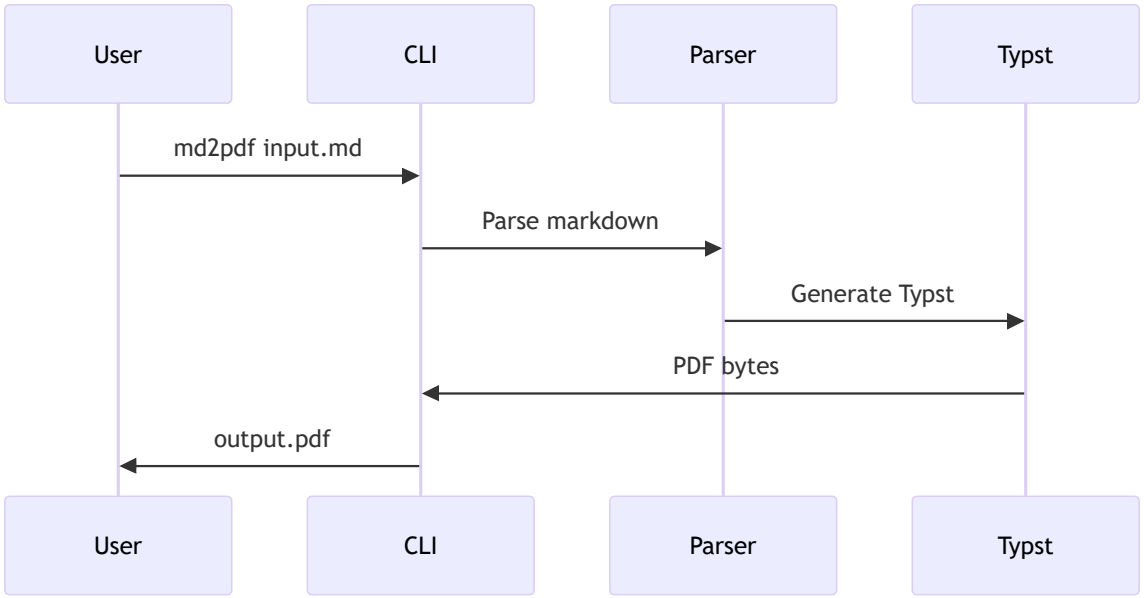
Mermaid Diagrams

Mermaid diagrams are rendered via kroki.io and embedded as SVG images.

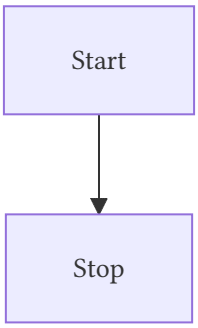
Flowchart



Sequence Diagram



Simple Flow



Callout Boxes

GitHub-Style Alerts

> 💡 **Tip**
>
> This is a tip callout. Use it to highlight helpful information.

> ⚠️ **Warning**
>
> This is a warning callout. Use it for potential issues.

> 📝 **Note**
>
> This is a note callout. Use it for additional context.

> ❗ **Important**
>
> This is an important callout. Use it for critical information.

> ⚠️ **Caution**
>
> This is a caution callout. Use it for dangerous operations.

Emoji-Style Callouts

Note: You can also use text-prefixed callouts for compatibility.

Blockquotes

> "Simplicity is the ultimate sophistication."This is a regular blockquote that spans multiple lines. It can contain various formatting like **bold** and *italic* text.

Horizontal Rules

Above the rule.

Below the rule.

Layout Directives

New Page

This content appears on a new page after the page break directive.

Two-Column Layout

Left Column Content

This text will appear in a two-column layout. Multi-column layouts are useful for comparison content or fitting more information on a page.

Right Column Content

The column layout automatically distributes content between the columns. Typst handles the column balancing.

Conclusion

The md2pdf Rust rewrite successfully implements the core features needed for markdown to PDF conversion:

1. **Complete Markdown Support**: All CommonMark features plus extensions
2. **Professional Output**: Cover pages, TOC, headers/footers
3. **Advanced Features**: Math equations, code highlighting, callouts
4. **Layout Control**: Page breaks, column layouts

Future Work

- Web API implementation (Phase 15)
- MCP Server for Claude integration (Phase 16)
- Google Fonts auto-download
- Image gallery support

Generated with md2pdf (Rust + Typst)